Fiscal Year:	FY 2007	**Task Last Updated:**	FY 10/29/2007
PI Name:	Lipa, John A. Ph.D.		
Project Title:	Universality Tests on DYNAMAX		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94305	**Congressional District:**	14
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	10/01/2003	**End Date:**	04/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 04/30/2007 per PI (1/07)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NAG 3 2873		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity.		

Task Description:	The purpose of this project was to provide scientific support for an improved test of the universality hypothesis in condensed matter physics using the DYNAMX (Dynamic Scaling Experiment) instrument that was being developed for deployment on the International Space Station. The DYNAMX apparatus was planned to be flown in the LTMPF (Low Temperature Microgravity Physics Facility) to measure the behavior of the thermal conductivity near the lambda point of helium. This quantity is of great theoretical interest for testing modern dynamical theories of phase transitions in condensed matter. Due to the cancellation of the FP discipline, work has been refocused on related ground measurements involving the specific heat and on better defining the flight measurements. Some possibility still exists for a free flier mission. The DYNAMX experiment provides access to the lambda transition of helium with the effects of gravity removed. This allows the testing of various aspects of phase transition theory with unprecedented accuracy. Our measurements would probe the behavior of the superfluid density and the specific heat very near the lambda lines to the highest level possible with present technology. The results of the measurements will be used to test well-established predictions the behavior of the heat capacity and the superfluid density very near the transition. This could become the highest quality test of modern theories of phase transitions in condensed matter.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Improved understanding of the applicability of renormalization techniques could affect the scientific infrastructure underlying many areas of practical value, from the limits on miniaturization of computer chips to the development of new types of polymers.
Task Progress:	The planned flight heat capacity measurements above and below the transition would span a narrow temperature range, nominally ± 2 mK wide. This work would be similar to LPE (Lambda point experiment) and aimed at improving our knowledge of the singularity in the bulk heat capacity at the transition. It would focus roughly equally on each side of the transition. It would be synergistic with the CQ (Specific heat at constant Q) experiment, providing wider range data at heat flux Q = 0, and benefiting from the software already developed for that experiment. The only additional requirements would be the ability to perform measurements over a wider temperature range with bigger heat pulses and to have good calibration of the temperature scale, hopefully to the 0.02% level or better. Also the pressure or 3He content of the sample would need to be changed. The DYNAMX project has developed state-of-the-art thermometry that would allow us to perform precision measurements of the specific heat, limited primarily by thermal relaxation effects. Because of the design of the cell, it would be possible to monitor the internal temperature of the helium at three locations as it relaxes towards equilibrium, giving us additional information to limit the measurement errors. The measurements would extend as close to the transition as possible to provide the maximum range of data for exponent estimation and to minimize the effects of the Wegner series and other corrections. We have simulated a data collection scenario below the transition lasting 4 days and find that it should be possible to obtain an uncertainty of a of < 0.0001. The simulation assumed that systematic effects could be controlled to 0.01% from pulse to pulse and that the thermometer noise was at the expected DYNAMX flight level. Measurements above the transition are harder to simulate accurately, but based on the short thermal relaxation time of the DYNAMX cell we would expect only a slight degradation in this region. An additional 4.4 days would be needed to cover this region adequately. An important output of this work would be to obtain a precise value of a below the transition in a range of temperature ten times narrower than in LPE. The reduction of the range decreases the sensitivity of the result to the leading Wegner series term in eq. 2 by about a factor of three, and the higher order terms by at least factor of ten. Thus the exponent would more closely reflect an asymptotic value, less perturbed by the confluent singularity. By restricting the temperature range even further, the sensitivity of the exponent to fitting range could be established with very good accuracy. Since confluent singularity terms would already be allowed for, such a result would go well beyond the 'effective exponent' testing of critical phenomena encountered in gas-liquid systems. Such a test of the basic functional form of a divergent property would probably be unique in the field of critical phenomena. A second result would be an improved measurement of the specific heat behavior above the transition, a region where LPE had significant difficulty. By combining such data with wider range ground measurements we expect to be able to bring our knowledge of the high temperature branch of the specific heat singularity to a level approaching that of the low temperature branch. This would give the best test to date of the predicted equality of the exponents on the two sides of the transition and would result in improved values for other universal quantities. Universality could then be tested by altering the pressure or 3He content of the sample using a valve already on the DYNAMX cell and a simple room temperature gas handling system. The major part of our laboratory research effort was focused on collecting new heat capacity data along the lambda line of helium to serve as precursor measurements for the potential flight experiment using the DYNAMX instrument. Accurate measurements of the specific heat of helium near the lambda line as a function of pressure are important for testing the predictions of universality that are a central feature of the theory of second order phase transitions. These predictions have been tested with moderate precision in a number of experiments and some small discrepancies have been noted. Over the period of the project we built a new apparatus for measuring of the constant-volume specific heat, Cv, in which the fluid pressure can also be recorded as a function of temperature. This information enables us to compute the constant-pressure specific heat, Cp, to which universality should apply. This information then allows us to compute the superfluid density from earlier second sound velocity measurements much more accurately than has been possible to date. The present results extend significantly closer to the transition than previous work, allowing a better determination of the asymptotic behavior. We were able to collect data and perform a preliminary analysis of the measurements using earlier pressure measurements obtained by others to convert from Cv to Cp. Our own pressure measurements are still being analyzed and are expected to result in significantly more accurate Cp data close to the transition.
Bibliography Type:	Description: (Last Updated: 11/02/2007)
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